

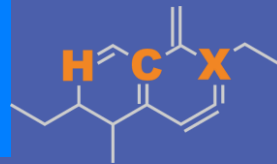
Ethanol Blending in Gasoline – El Salvador

July 2026



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Ethanol Blending in Latin America

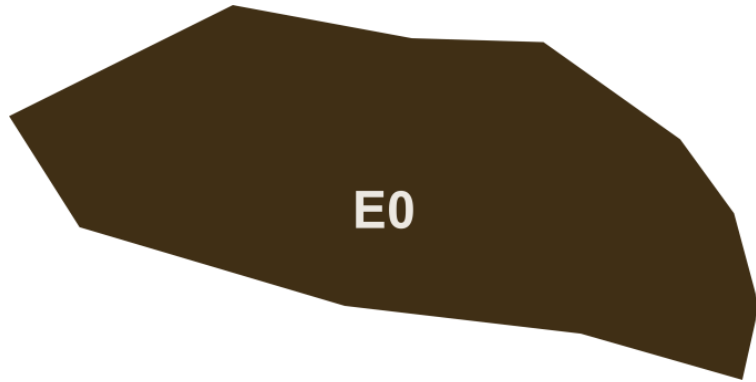
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



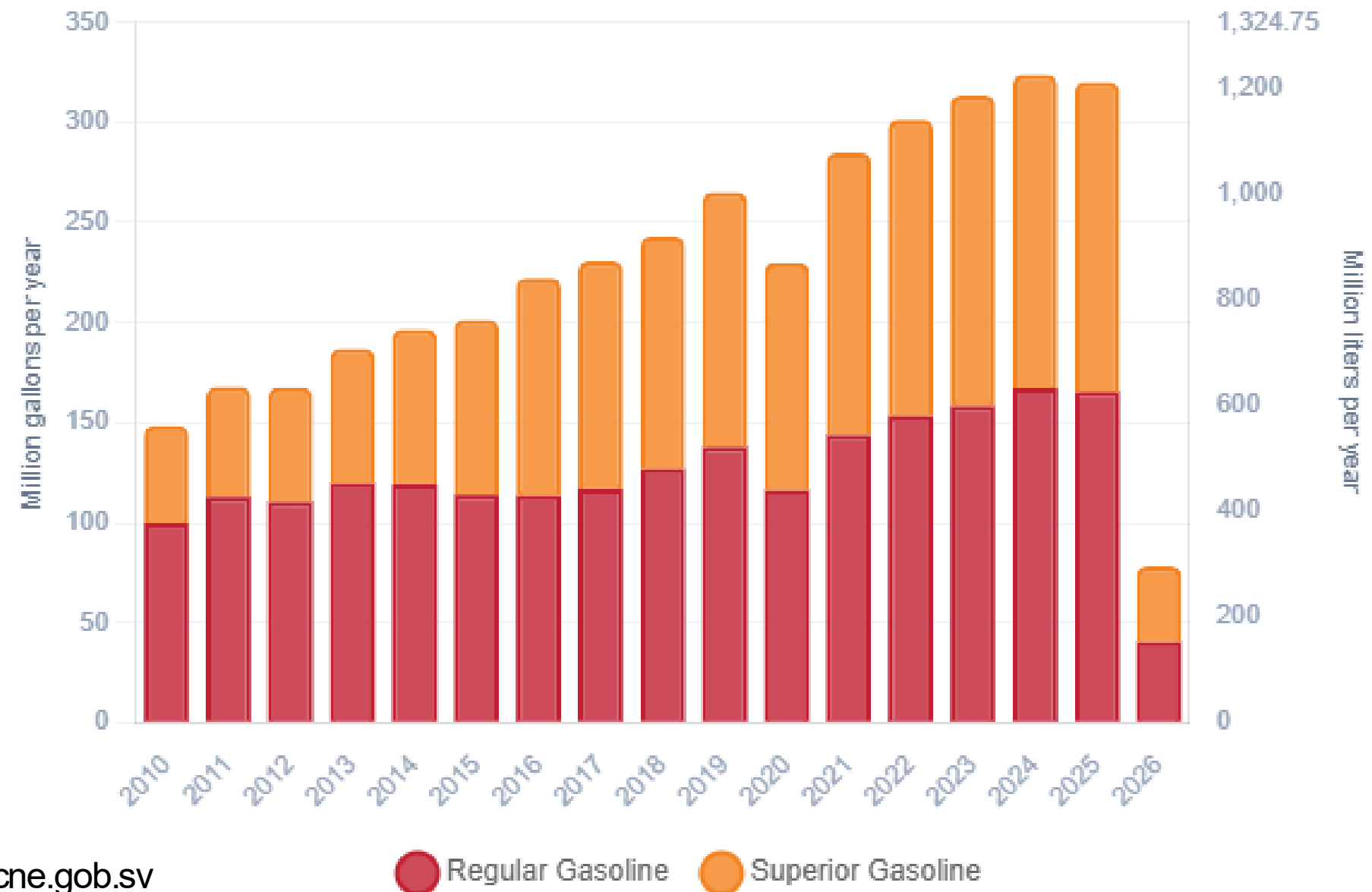
Ethanol Blending in Gasoline - El Salvador



In 2025, gasoline demand was 328 million gallons (1,241 million liters). Regular gasoline RON 91 represented 51% of the volume consumed while Superior gasoline RON 95 reached 49%. Gasoline is supplied only by imports, distributed mainly by Chevron, Puma and Uno companies and imported mainly from the United States.

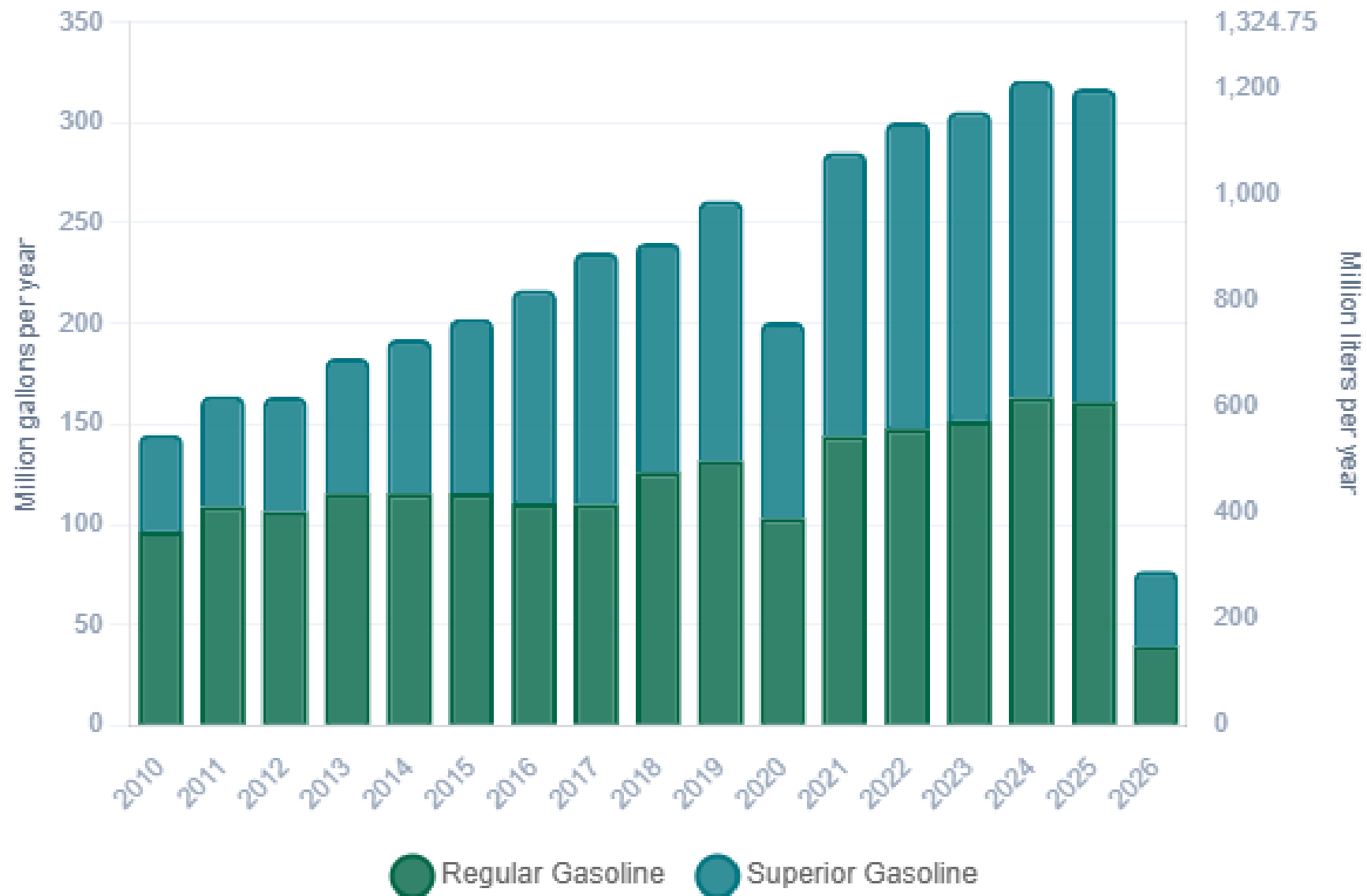
El Salvador has no national mandate for ethanol. It has the potential to produce ethanol from maize and sugar cane. In the last year there has been an interest for its use.

Gasoline Demand in El Salvador



Source: cne.gob.sv

Gasoline Imports in El Salvador



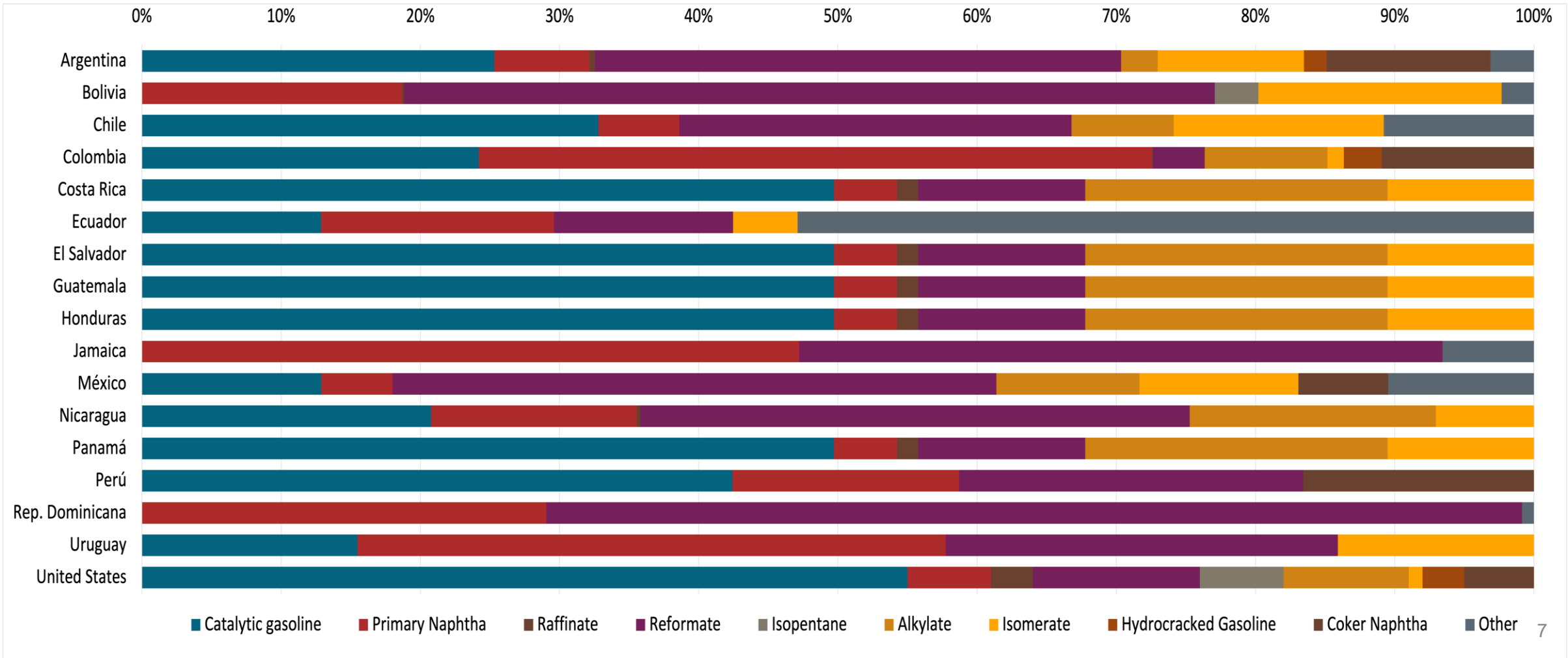
Gasoline Quality in El Salvador

Name	RTCA 75.01.19:19	RTCA 75.01.20:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	5% v/v	5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	2,0% v/v	2,0% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	0,7% v/v	0,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: RICA

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



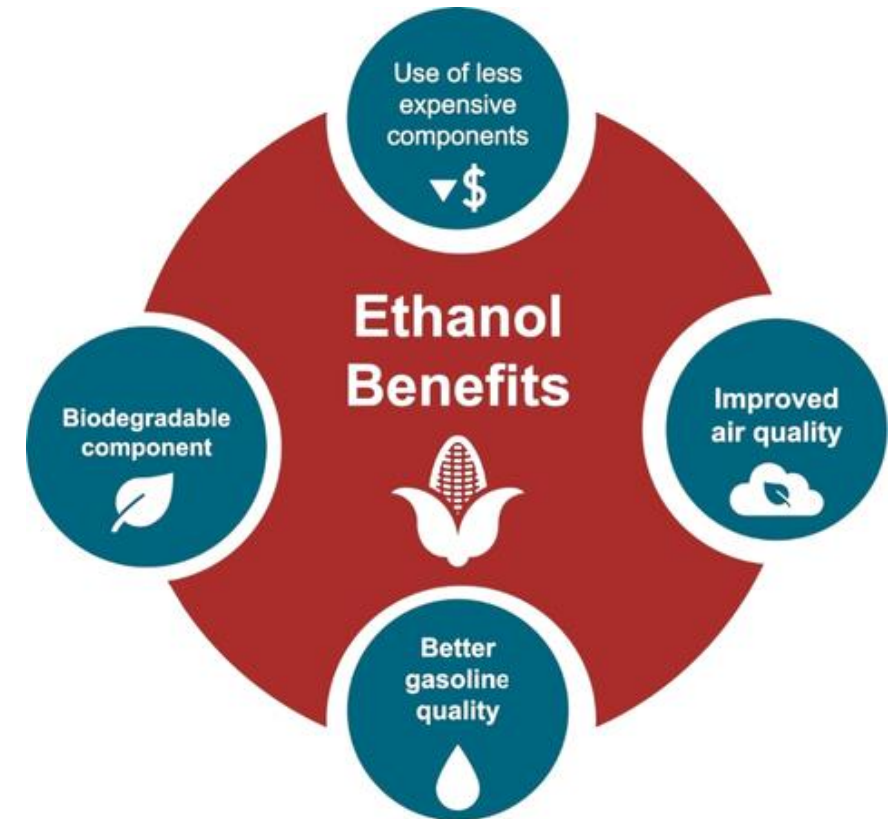
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

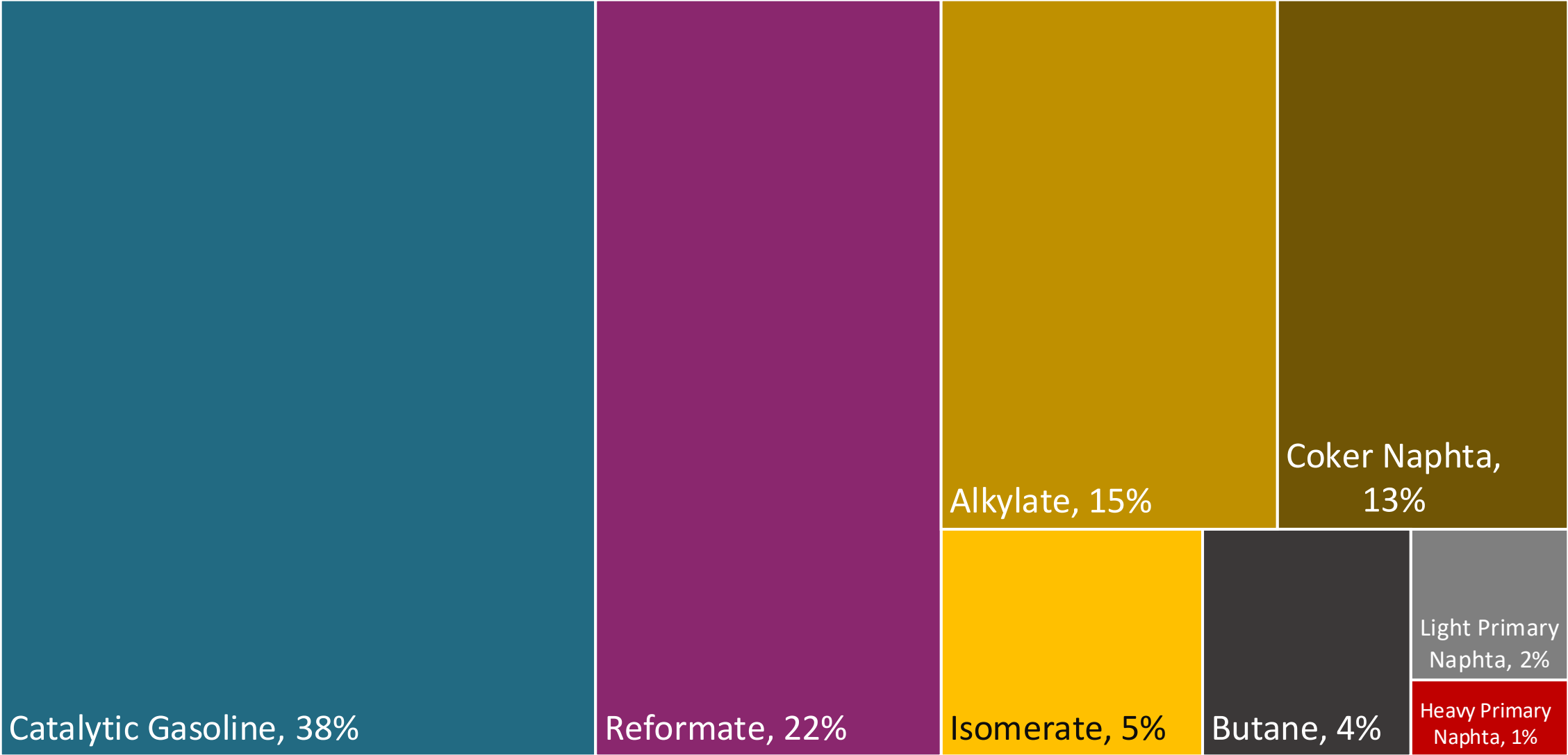
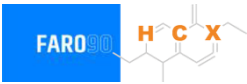
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

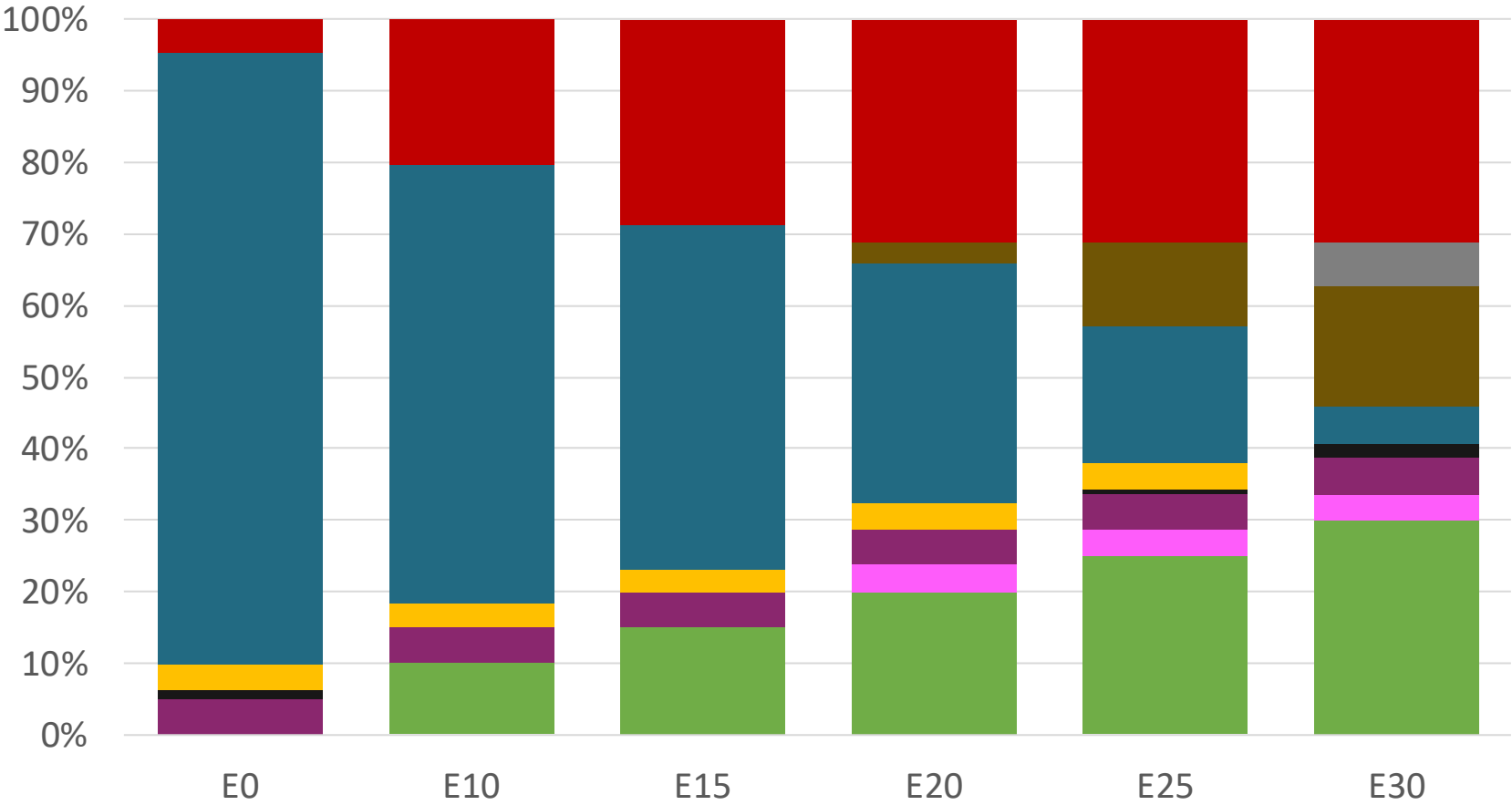
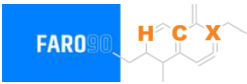


Blending Components- El Salvador



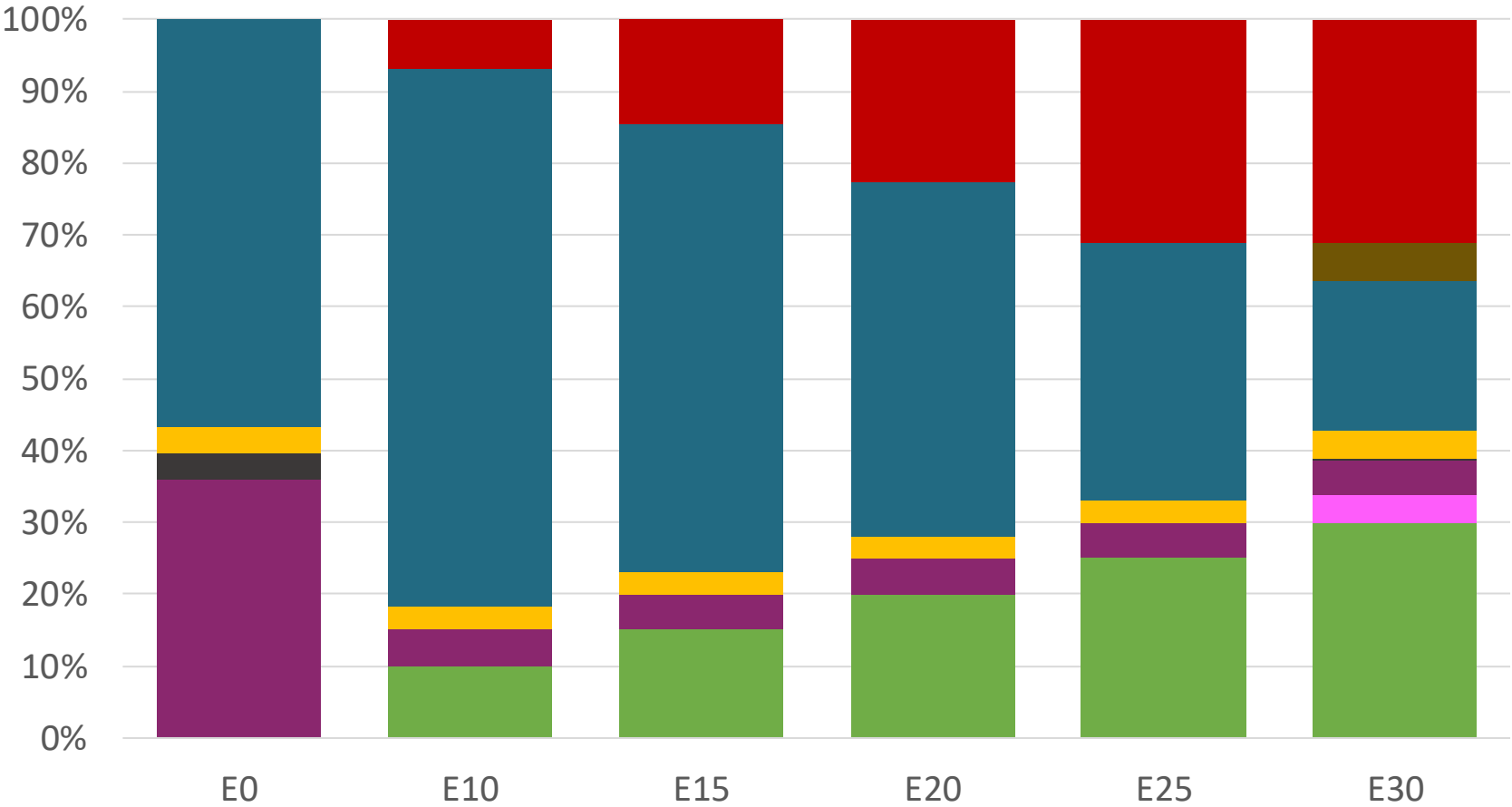
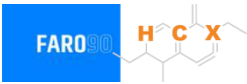
Source: Faro90

EI Salvador – Regular – Constant Octane



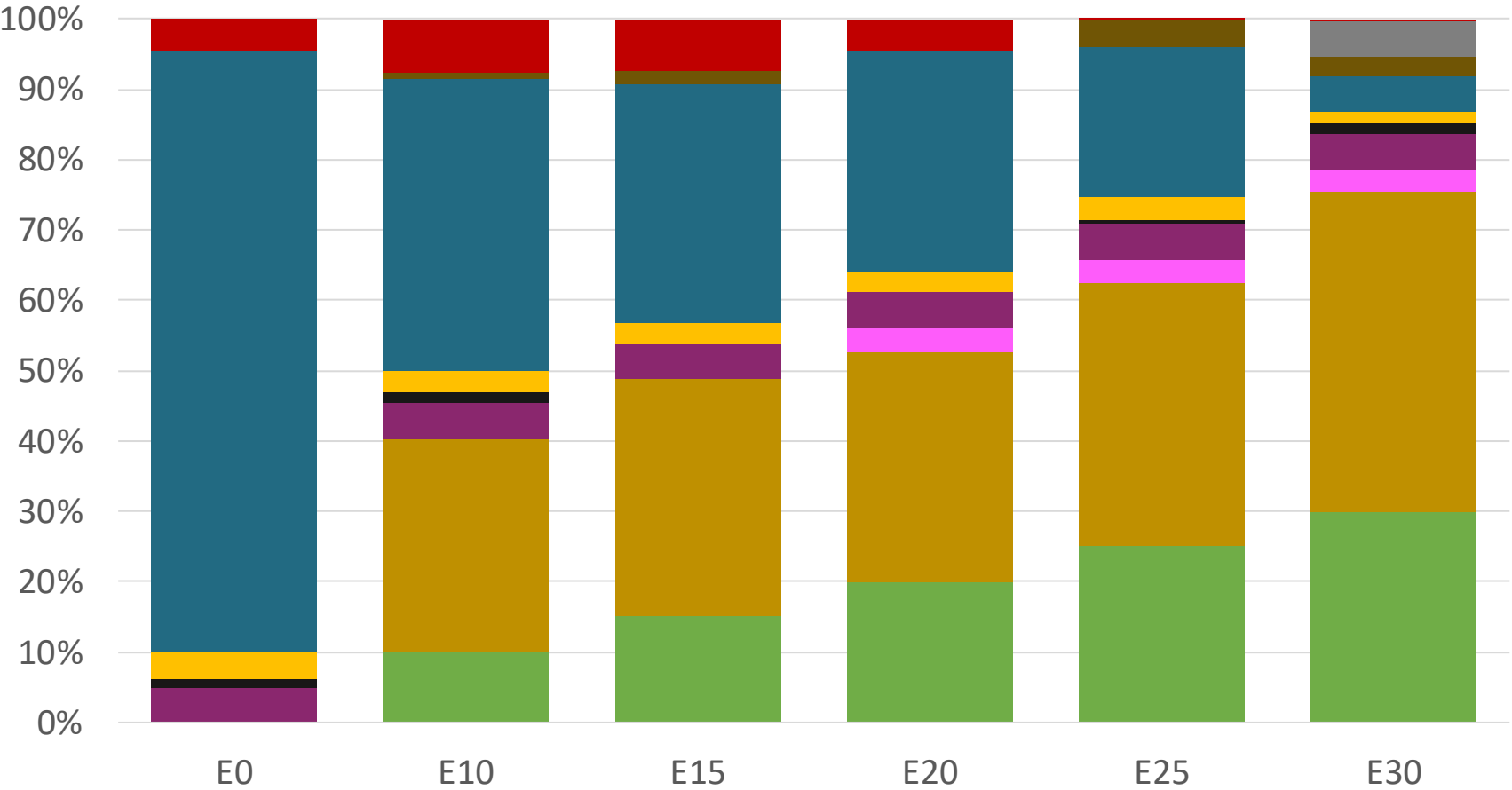
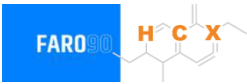
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.126	\$. 1.867	\$. 1.723	\$. 1.576	\$ 1.425	\$. 1.307

El Salvador – Premium – Constant Octane



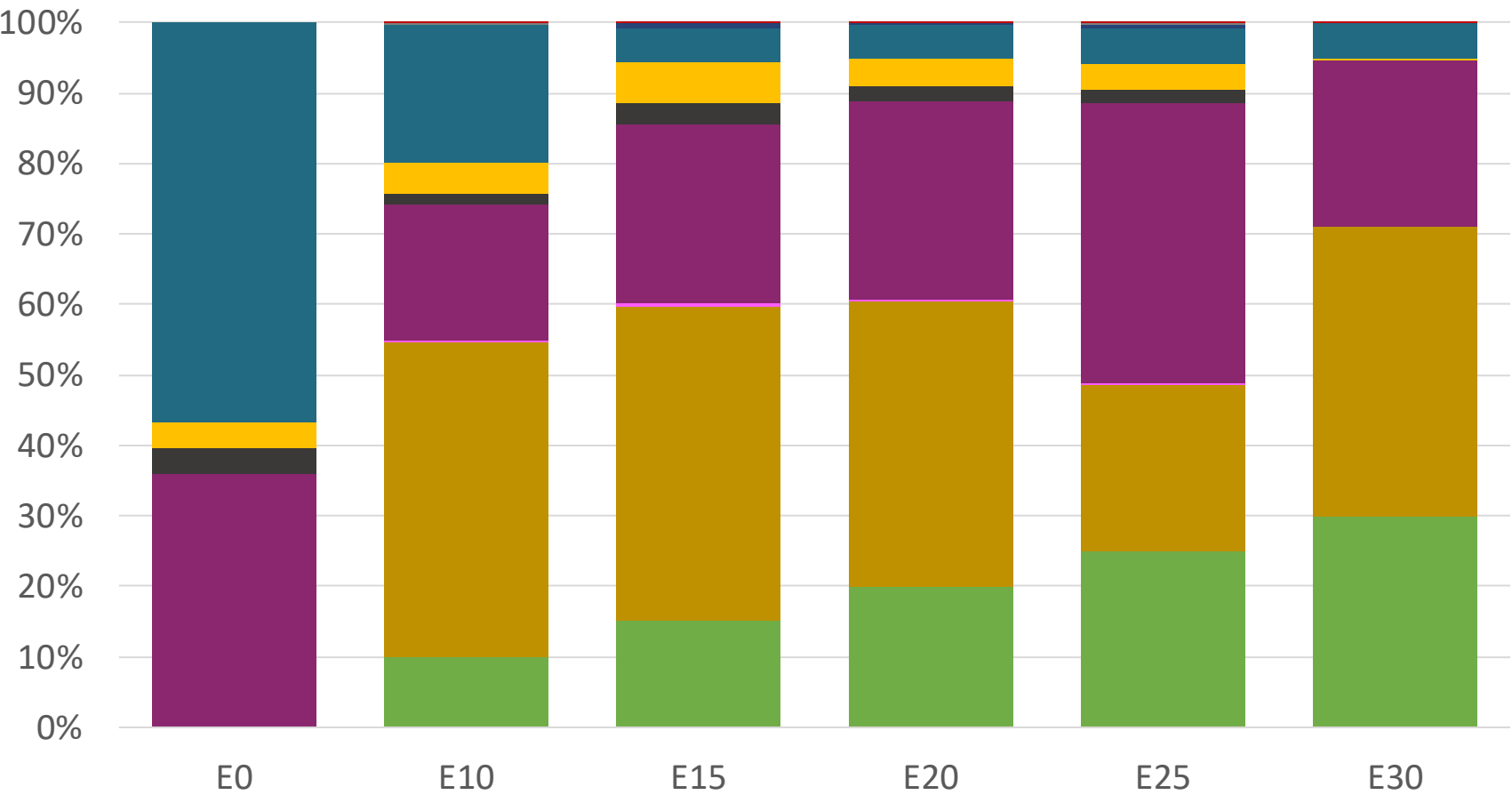
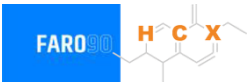
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.376	\$ 2.073	\$ 1.938	\$ 1.796	\$ 1.647	\$ 1.498

El Salvador – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126

El Salvador – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

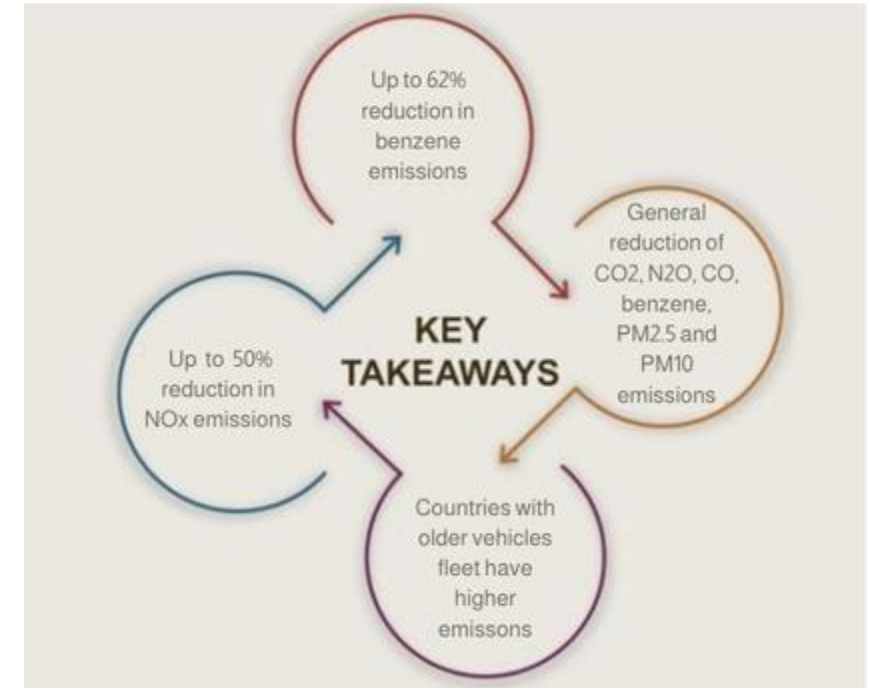
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicular Fleet in El Salvador

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles			
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	5			
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	739			
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	15,239			
		Euro III		> 8 years, > 160 tkm	37,987			
		Euro IV		> 8 years, > 160 tkm	95,925			
		Euro V / Euro 6		> 8 years, > 160 tkm	143,401			
				> 8 years, > 160 tkm	178,359			
				4-8 years, 80-160 tkm	167,508			
				< 4 years, < 80 tkm				
	Hybrid			Any	152			
Electric			Any	119				
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	12			
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	1,239			
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	17,096			
		Euro III		> 8 years, > 160 tkm	32,828			
		Euro IV		> 8 years, > 160 tkm	64,441			
		Euro V / Euro 6		> 8 years, > 160 tkm	73,656			
				> 8 years, > 160 tkm	67,493			
				4-8 years, 80-160 tkm	54,893			
				< 4 years, < 80 tkm				
	Hybrid			Any	74			
Electric			Any	58				
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	3,788			
		Euro II		> 8 years, > 160 tkm	16,390			
		Euro III		> 8 years, > 160 tkm	42,231			
		Euro IV		> 8 years, > 160 tkm	26,009			
		Euro V		4-8 years, 80-160 tkm	122,044			
				4-8 years, 80-160 tkm	250,554			
			< 4 years, < 80 tkm	241,102				
	Hybrid			Any	154			
	Electric			Any	120			
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	1,093			
	FI	3-Ways		> 8 years, > 160 tkm	24,117			
		Euro III		> 8 years, > 160 tkm	21,929			
		Euro IV		> 8 years, > 160 tkm	22,379			
		Euro V / Euro 6		> 8 years, > 160 tkm	16,188			
				4-8 years, 80-160 tkm	26,181			
				< 4 years, < 80 tkm	114,901			
	Hybrid			Any	20			
	Electric			Any	16			

Vehicular Fleet: 1,652,936

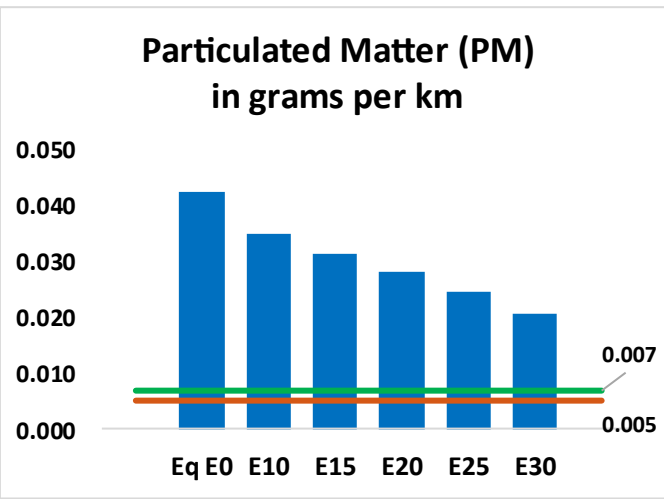
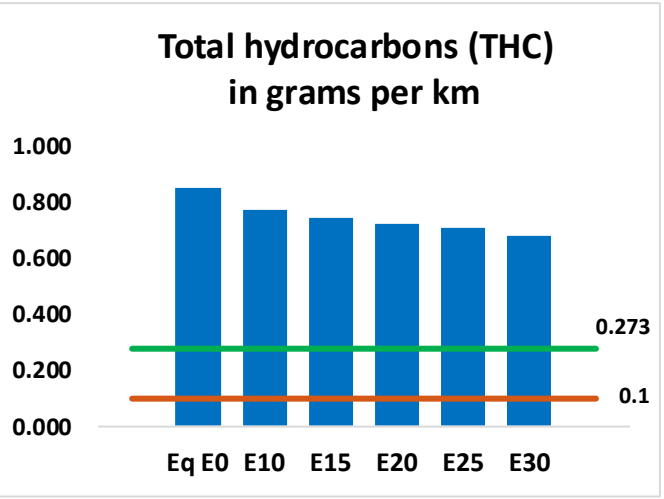
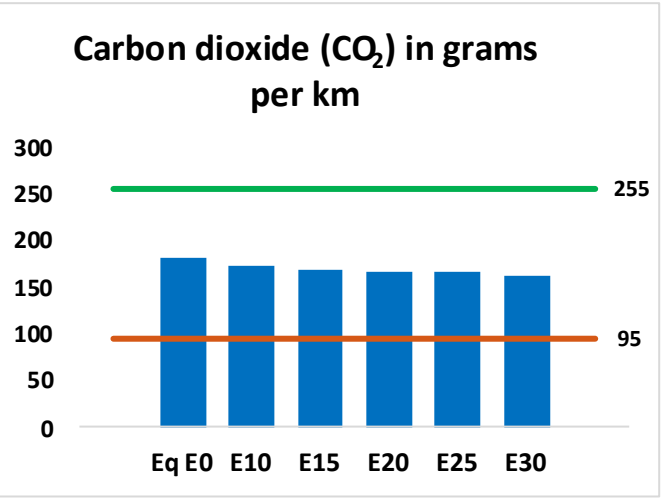
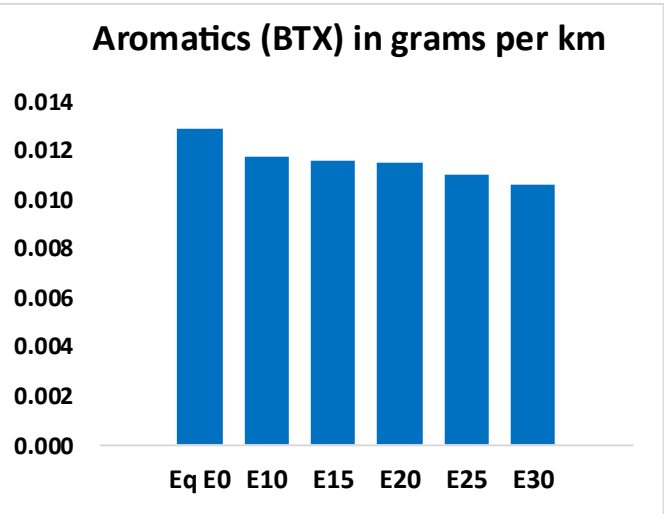
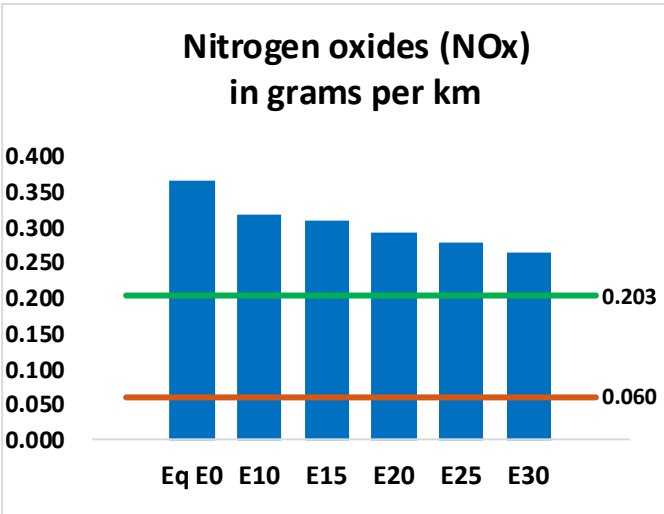
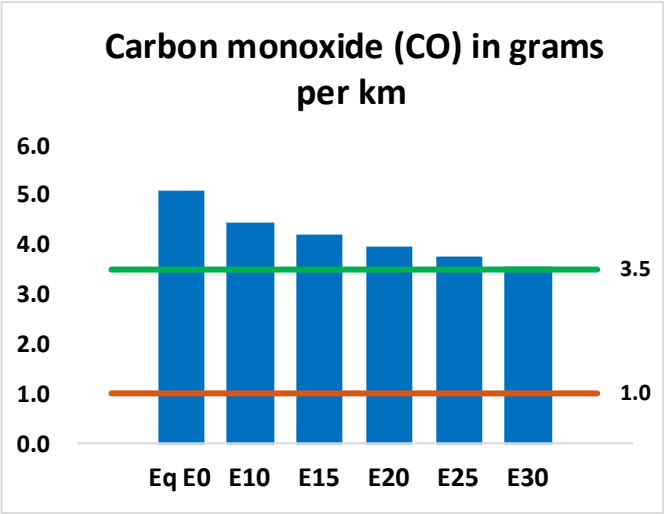
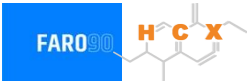
Average Age: 9 years

Motorcycles: 33%

Source: Registro Publico, Faro 90

El Salvador - Vehicular Emissions

United States
Euro 6



El Salvador - Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	164,291	154,054	143,818	135,896	128,140	122,240	115,143	-12%	-22%
CO2	5,847,536	5,697,171	5,555,160	5,443,472	5,388,204	5,364,902	5,266,011	-5%	-8%
VOC	27,515	26,272	25,029	24,181	23,406	22,833	21,985	-9%	-15%
NOx	11,842	10,934	10,304	9,979	9,460	9,051	8,570	-13%	-20%
SOx	67	62	57	52	48	44	39	-15%	-28%
NH3	2,277	2,254	2,232	2,242	2,267	2,285	2,300	-2%	0%
N2O	71	71	70	71	72	73	74	-1%	2%
CH4	6,419	6,419	6,419	6,515	6,656	6,778	6,894	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	139	130	121	115	108	103	96	-12%	-22%
Acetaldehyde	385	442	649	988	1,346	1,538	1,817	68%	249%
Formaldehyde	1,520	1,478	1,723	2,023	2,120	2,345	2,553	13%	39%
Benzene	418	402	381	375	372	356	344	-9%	-11%
PM2.5	359	326	293	265	237	207	175	-18%	-34%
PM10	1,018	925	833	751	672	586	497	-18%	-34%

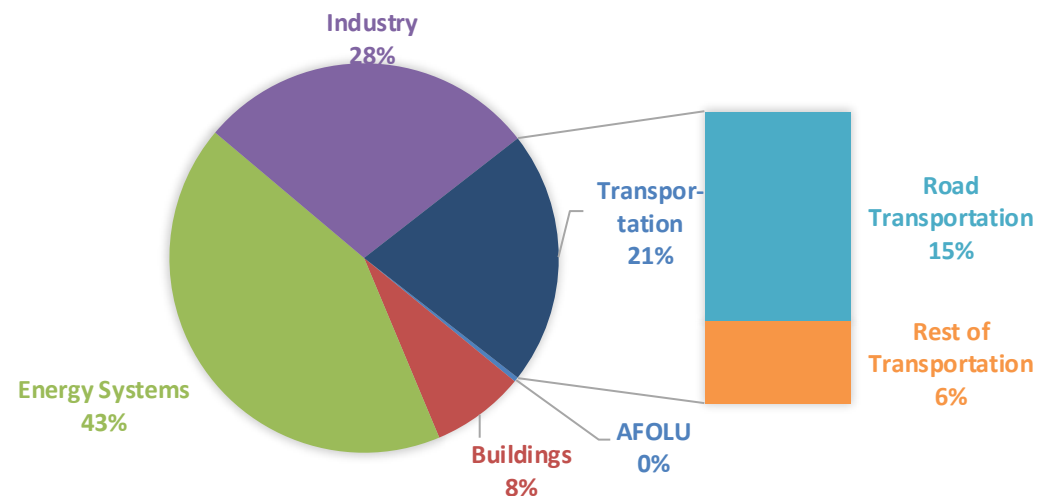
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

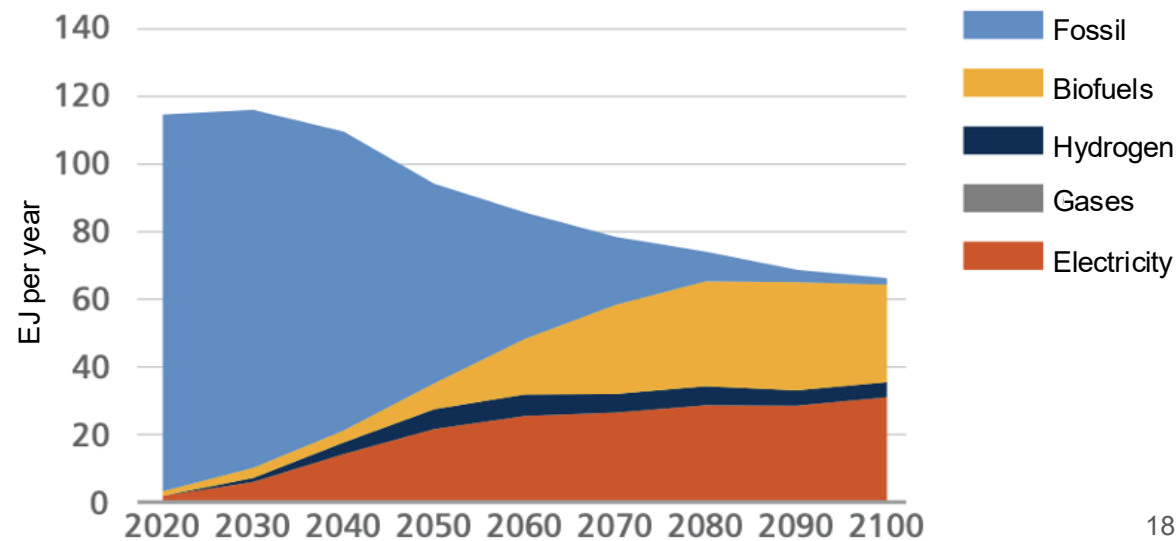
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

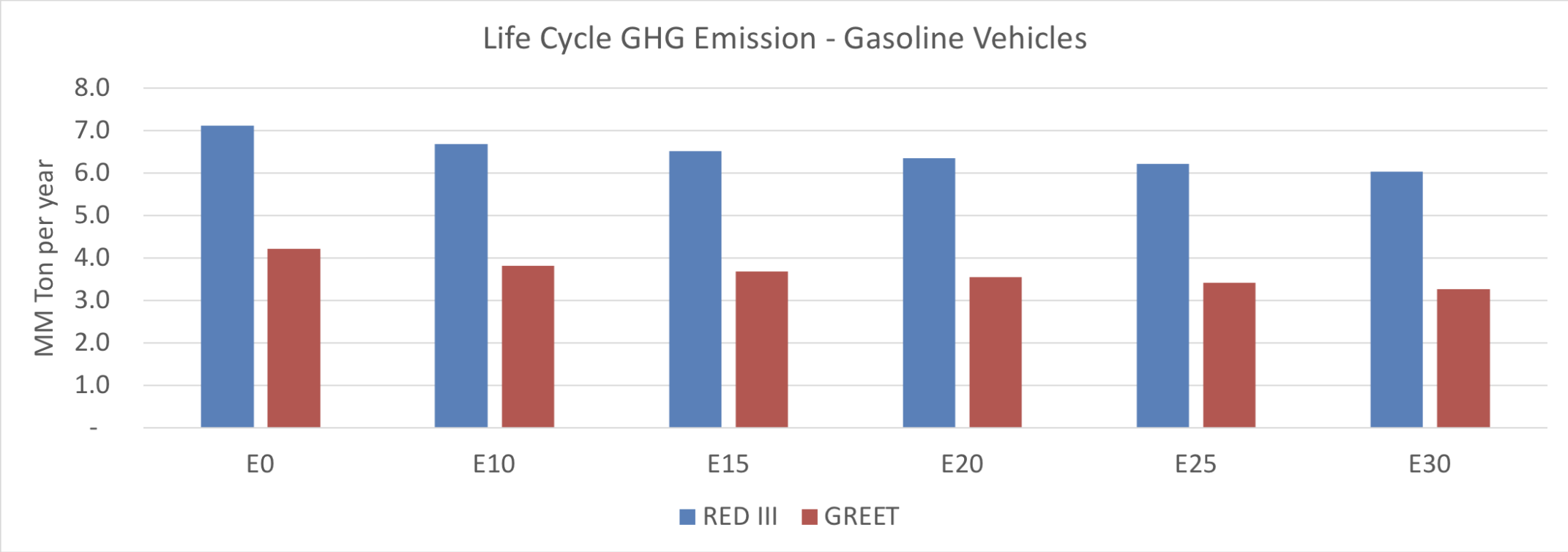
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



El Salvador – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	14.8
2035 Target (MMT/year)	10.3
Delta	4.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.4%	12.7%	16.0%	18.8%	22.6%
RED II/III	0.0%	6.3%	8.6%	10.8%	12.8%	15.2%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	8.9%	12.0%	15.2%	17.9%	21.5%
RED II/III	0.0%	10.1%	13.8%	17.4%	20.6%	24.5%